

AKKALA VENKATA SAI KRISHNA

Machine Learning Engineer | Data Scientist | Generative AI Developer

Bangalore, India • +91 9676295867 • Gmail : saikrishnaakkala9@gmail.com • [LinkedIn](#) • [GitHub](#)

PROFESSIONAL SUMMARY

AI Engineer with hands-on experience building AI-powered solutions that drive real business impact. Skilled at turning complex data into clear insights and deploying intelligent systems across cloud infrastructure. Comfortable working independently and collaborating with both technical and non-technical stakeholders.

TECHNICAL SKILLS

Languages:	Python, SQL, Advanced Excel
ML / AI:	Scikit-learn, Statistical Modelling, Machine Learning Models, NLP, Deep Learning, Predictive Modelling
Generative AI:	Generative AI (RAG Pipelines), LangChain, Prompt Engineering, Embedding Generation, Vector Databases, Foundation Models, LLMs
Cloud & AWS:	Amazon Bedrock, AWS OpenSearch (Vector DB), Amazon EC2, AWS S3
Data Science:	EDA, Feature Engineering, Data Mining, Statistical Analysis, Hypothesis Testing, Data Modelling, ETL Pipelines, Star Schema
Visualization:	Power BI, Tableau, Matplotlib, Seaborn, Streamlit
Databases:	Amazon OpenSearch, PostgreSQL, SQL Server
Tools:	Hugging Face Transformers, Pandas, NumPy, GitHub, Jupyter Notebook, VS Code

PROFESSIONAL EXPERIENCE

AI Intern | Hitachi Digital Services | Bangalore

Jan 2025 – Present

Project: Multimodal RAG-Based Conversational AI Chatbot (Client: A Japan Car Company)

- Partnered with cross-functional stakeholders to identify high-impact applications of data; designed and architected a production-grade Multimodal RAG pipeline using LangChain and Amazon Bedrock, cutting manual documentation search time by ~60%
- Mined and analyzed 10,000+ document chunks; built semantic vector search on Amazon OpenSearch with sub-second similarity retrieval for large-scale enterprise datasets
- Developed and refined machine learning models and NLP-based retrieval systems; integrated Claude and Titan LLMs for context-aware, self-learning response generation
- Engineered prompt templates and embedding tuning (chunk size, overlap), applying statistical analysis to boost retrieval precision by ~35% via iterative A/B evaluation
- Deployed full-stack AI backend on Amazon EC2 with REST API endpoints, enabling scalable, concurrent enterprise query handling
- Evaluated model performance with custom test suites; iteratively improved contextual coherence, reduced hallucinations, and automated model quality checks
- Provided data quality feedback loops — cross-checking sources, validating data correlation, and ensuring adequacy and legitimacy of ingested datasets

PROJECTS

AI-Powered Jira Resolution Agent CrewAI | Ollama | Python | Git

- ▶ Built an end-to-end agentic AI system that autonomously reads Jira tickets, identifies root causes of code-level bugs, generates targeted fixes, and commits changes to Git branches — eliminating manual developer context-switching
- ▶ Designed a multi-step LLM reasoning pipeline that chains ticket description, stack traces, and relevant source files to enable root cause analysis, going beyond surface-level error detection
- ▶ Orchestrated AI agents using CrewAI with automated Git workflows, enabling full traceability through structured logs across every stage of the resolution lifecycle
- ▶ Reduced manual effort in the bug-fix cycle by integrating Jira, codebase analysis, and version control into a single intelligent workflow, demonstrating applied skills in LLM orchestration and agentic system design

Enterprise Data Warehouse & Analytics Platform Python | SQL | ETL | Star Schema | Power BI | Statistical Modelling

- ▶ Identified high-impact business problems and designed end-to-end ETL pipeline ingesting CRM and ERP datasets; applied transformation logic and data quality checks, enabling 3× faster reporting queries
- ▶ Performed exploratory data analysis (EDA) to discover sales trends, customer segmentation patterns, and operational insights — communicating findings to both technical and non-technical stakeholders
- ▶ Modeled Star Schema with optimized fact and dimension tables; developed statistical models for sales trend analysis and customer segmentation supporting meaningful business decisions
- ▶ Integrated with Power BI and Tableau for self-service analytics dashboards, reducing time-to-insight and enabling business stakeholders to independently query data

Automated EDA & Intelligent Data Analysis Platform Python | Streamlit | Pandas | SQL Server | Machine Learning

- ▶ Built a Streamlit web application for CSV and SQL Server data ingestion with automated EDA, statistical profiling, and machine learning-based outlier detection — reducing analyst data prep time by ~50%
- ▶ Implemented automated data cleaning, dynamic Matplotlib/Seaborn visualizations, and interactive filtering to unearth insights from datasets up to 500K rows
- ▶ Enabled non-technical users to perform exploratory data analysis independently, communicating complex statistics through intuitive, interactive dashboards
- ▶ Designed and A/B tested different data processing pipelines to evaluate model performance and optimize statistical output quality

EDUCATION

B.Tech — Data Science

Santhiram Engineering College (JNTUA)

CGPA: 8.49 / 10

Aug 2021 – Apr 2025

CERTIFICATIONS

- ▶ Data Analyst Certification - ExcelR, Bangalore
- ▶ Creating Agents with CrewAI - LinkedIn
- ▶ NLP — LinkedIn
- ▶ Advanced Prompt Engineering Techniques - LinkedIn